Molave Liggayu

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TECHNICAL SKILLS

Languages: C, C++, ARM assembly, Java, JavaScript, HTML / CSS, Python, SQL, Bash
Tools & Frameworks: Arduino/STM toolchains, Git, Docker, GDB, Gradle, Playwright, Visual Studio, Wireshark
Operating Systems: Windows, UNIX/Linux
Simulation & Design: Quartus, ModelSim, MATLAB, SOLIDWORKS

EDUCATION

University of British Columbia
Bachelor of Applied Science - Computer Engineering (3rd Year) **Expected Graduation:** ***May 2027***

- **CGPA:** 4.0 - Dean's List
- **Course Coverage:** Computer Architecture, Embedded Systems, Circuit Analysis, Operating Systems, Algorithm Design, Communication Protocols, More Here
- **Awards:** EGBC Award (2024), Renchko Family Scholarship (2023), Rutherford Scholarship (2023)

EXPERIENCE

Firmware Developer, Sailbot Design Team (Vancouver, BC) ***Sep 2024 – Present***

- Leading firmware development for power subsystems of POLARIS, an autonomous solar-powered research vessel monitoring marine health
- Programmed STM32U5 firmware and a custom bootloader enabling remote over-water updates via FDCAN/CAN, cutting maintenance time by 50% and minimizing manual servicing
- Integrated BQ76920-based BMS over I2C to report real-time current and voltage telemetry
- Engineered low-pass filters to suppress power-line voltage spikes and reduced supply noise by $\sim$ 20% on critical actuators

Student Intern Developer, atVenu (Calgary, AB) ***Jul 2022 – Sep 2022***

- Automated browser-based testing with Playwright, ensured cross-platform reliability across 5 browsers and 10+ devices
- Built an internal tool to validate inventory sync states, reducing user-side data discrepancies and improving reliability of live sales reports
- Practiced Agile and SOLID principles under senior developer mentorship in a team environment

PROJECTS

Turing complete RISC based CPU

- Designed and synthesized a 16-bit pipelined RISC CPU on DE1-SoC FPGA using SystemVerilog and Quartus toolchain
- Supports a 20-instruction ISA, tested through gate-level simulation and testbenches using the ModelSim toolchain

ML-based Motion Tracking app (NASA APP 2025) https://github.com/Re-Emzee/SpaceDance

- Achieved regional win and global nomination at the 2025 NASA Space Apps Hackathon
- Co-developed a Java Swing game linked to a Python/OpenCV motion-tracking system, enabling gesture-based interaction to support fitness in zero-gravity conditions

UNIX-Based Signal-Handling Shell Implementation

- Built a UNIX shell in C supporting concurrent execution for 32 processes with signal and interrupt driven control
- Implemented robust signal handlers for SIGINT, SIGTSTP, and custom process lifecycle management

Virtual Memory System and Dynamic Memory Allocator

- Designed a user-level virtual memory manager featuring a two-level hierarchical page table structure to simulate OS-style address translation for user processes
- Created a dynamic memory allocator (malloc/free implementations) and validated with stress tests on fragmented heaps

See more projects and portfolio here